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SYNTHESIS

# The dimensionality and structure of species trait spaces

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## Abstract

Trait-based ecology aims to understand the processes that generate the overarching diversity of organismal traits and their influence on ecosystem functioning. Achieving this goal requires simplifying this complexity in synthetic axes defining a trait space and to cluster species based on their traits while identifying those with unique combinations of traits. However, so far, we know little about the dimensionality, the robustness to trait omission and the structure of these trait spaces. Here, we propose a unified framework and a synthesis across 30 trait datasets representing a broad variety of taxa, ecosystems and spatial scales to show that a common trade-off between trait space quality and operationality appears between three and six dimensions. The robustness to trait omission is generally low but highly variable among datasets. We also highlight invariant scaling relationships, whatever organismal complexity, between the number of clusters, the number of species in the dominant cluster and the number of unique species with total species richness. When species richness increases, the number of unique species saturates, whereas species tend to disproportionately pack in the richest cluster. Based on these results, we propose some rules of thumb to build species trait spaces and estimate subsequent functional diversity indices.

David Mouillot and Nicolas Loiseau are co-first authors.

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**KEYWORDS**
complexity, functional ecology, hypervolume, species clustering, species uniqueness

## INTRODUCTION

Biodiversity comprises a great variety of organismal forms, functions, diets, physiologies and life histories—hereafter called traits—that have been shaped by large-scale evolutionary and ecological processes (Reich et al., 1999; Schluter, 1993) and that have important implications for ecosystem functioning (Duffy et al., 2001; Hector et al., 1999). Thus, quantifying and characterising trait variation among species is key to understand species assembly rules (Bruelheide et al., 2018; Jarzyna et al., 2020), evolutionary dynamics (Deline et al., 2018; Pigot et al., 2020) and ecosystem functioning (Cadotte, 2017; Gagic et al., 2015) but also to predict biodiversity responses to global changes (McLean et al., 2019; Rüger et al., 2020) and to guide conservation efforts (Pollock et al., 2017; Sala et al., 2021). For instance, experiments show that plant communities with higher levels of trait diversity are more productive and have a higher resource use efficiency by intercepting more light, taking up more nitrogen and occupying more of the available space (Spehn et al., 2005) but can also limit plant disease risks (Le Bagousse-Pinguet et al., 2021).

Yet, owing to the increasing availability of widespread—but also incomplete and heterogeneous—information on multiple traits collected with various methods across most kingdoms of life (Jones et al., 2009; Kattge et al., 2020; Perez et al., 2019; Schneider et al., 2017), the characterisation of species ecological strategies and relationships with environmental conditions is becoming more complex and multidimensional than ever (Bruelheide et al., 2018; Villeger et al., 2011). Reducing this complexity has both theoretical and practical benefits. First, clustering thousands of species into a limited number of entities sharing similar trait values can reveal the amount of functional vulnerability within assemblages (Mouillot et al., 2014) or a functional backbone common to separate geographic realms (McLean et al., 2021). Second, many traits are strongly correlated owing to life-history trade-offs or adaptive constraints, suggesting that trait diversity within a clade is more limited than expected (Díaz et al., 2016; Pigot et al., 2020; Winemiller et al., 2015). Birds with relatively long, narrow wings, pointed tips and strong sweep back (such as those of a swallow) fly at high speeds but are energetically inefficient and cannot fly over long distances (Savile, 1957). Third, the hyper-dimensionality of trait spaces, where species are placed according to their combinations of traits, prevents the computation of hypervolume-based functional diversity indices or null models to test community assembly hypotheses (Blonder et al., 2014; Maire et al., 2015). Fourth, predicting biodiversity and ecosystem trajectories under various environmental scenarios needs parsimonious trait-based models (Barros et al., 2017; Cooke et al., 2019b; Rüger et al., 2020) because the use of too many traits may induce overfitting (Bernhardt-Römermann et al., 2008).

However, we still lack a unified methodological framework to assess the different aspects of a species trait space. The dimensionality and the structure of a species trait space are indeed two sides of the same coin because they both refer to its complexity, that is, the way species and their traits are organised in this space. We also lack a synthesis on the main factors shaping the different aspects of species trait spaces. The degree of organismal complexity, which is related to the diversity of cell types (Valentine et al., 1994), can indeed influence the complexity of species trait space following key functional innovation in multicellular clades (Cox et al., 2021; Knoll, 2011; Sosiak & Barden, 2021). The environment can also be crucial in determining the course of multicellular evolution and organismal complexity, with aggregative multicellularity evolving more frequently on land whereas clonal multicellularity is more frequent in water (Fisher et al., 2020). On the other hand, the number of species and trait characteristics are likely to influence the complexity of species trait spaces beyond the type of organism and the environment (Kohli & Jarzyna, 2021; Zhu et al., 2017). Yet the relative importance of these different potential drivers has never been tested across kingdoms and realms for a vast number and diversity of traits and taxa.

A first critical aspect of a species trait space refers to the well-known dimensionality issue (Laughlin, 2014; Maire et al., 2015). While dimension reduction is appealing, the devil lies in the details. Indeed, going from a large number of traits to a reduced trait space (Figure 1a–d), that represents meaningful ecological dimensions or axes, is conceptually and methodologically difficult (Maire et al., 2015; Pigot et al., 2020; Sosiak & Barden, 2021; Winemiller et al., 2015). High-dimensional spaces might indeed be required to fully capture trait variation among species (Carscadden et al., 2017) or clades (Cooney et al., 2017). Moreover, the extent to which collected traits, some being potentially uninformative, redundant or incomplete, can be summarised with a few dimensions to reliably represent the diversity of organism forms and functions has not been quantitatively tested across a large set of taxa, ecosystems and traits.

A second key aspect of any species trait space is its robustness to the choice or the omission of traits so its capacity to consistently position species relative to each other whatever the sub-selection of traits for a given goal (environmental filtering, competitive interactions etc.). This capacity ultimately determines the confidence by which we can estimate metrics like species trait dissimilarity or functional diversity (Carscadden et al., 2017;

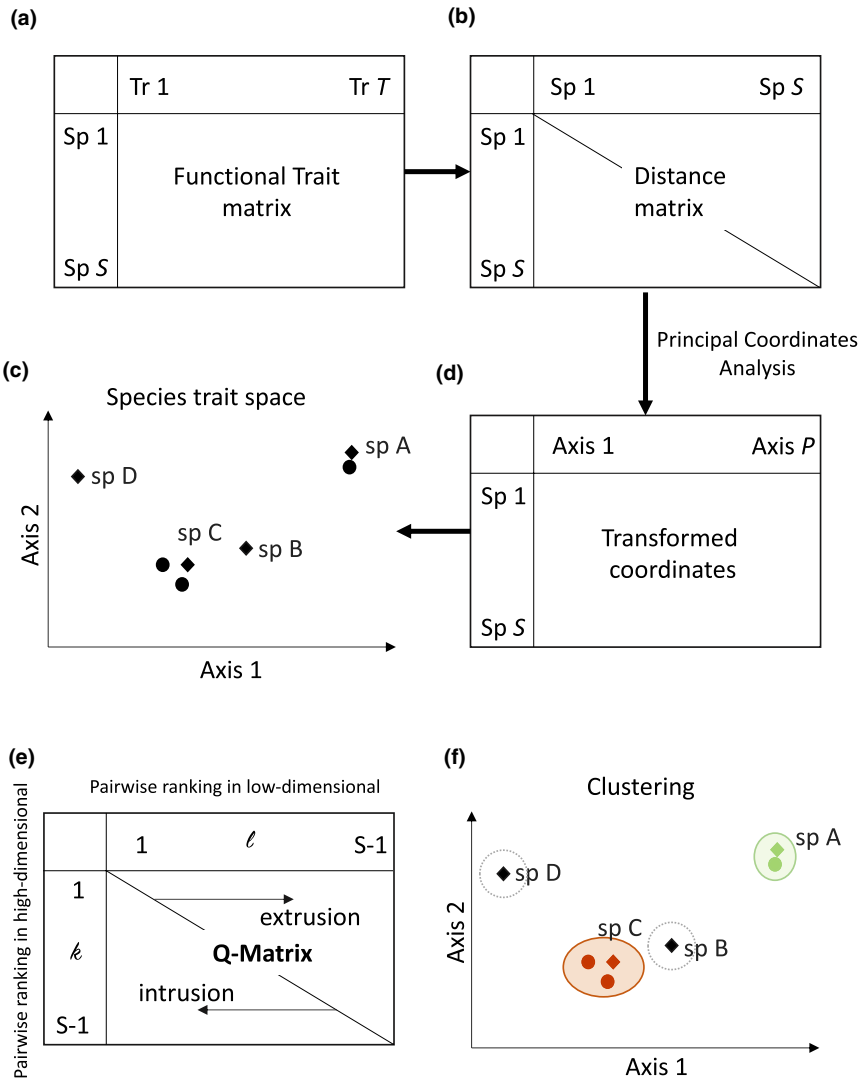



**FIGURE 1** Theoretical example showing the different steps of our framework from species trait matrix (a) to species trait space (d) after calculating species pairwise distances (b) and extracting synthetic axes providing new species coordinates in a low-dimensional space (c). Then the ranking of species pairs in both high-dimensional (i.e., considering all traits so distance matrix (b)) and low-dimensional spaces (i.e., considering coordinates on few axes in (c)) can provide a $Q$ matrix where the diagonal corresponds to all species pairs with a perfect match in their ranking in both spaces whereas off-diagonal values correspond to mismatching species pairs in the co-ranking; that is, species get closer in the low-dimensional space (intrusion) or farther (extrusion) compared with their relative position in the high-dimensional space. A clustering algorithm isolates two unique species (species B and D) in the trait space (no neighbours within a given radius $d_0$) and creates two clusters with two (green) and three (red) species (f). See Section 2 for details

Kohli & Jarzyna, 2021; Zhu et al., 2017). However, this robustness has been largely overlooked and deserves a dedicated analysis across multiple datasets where the number, completeness, correlation and type of traits cover a broad range of options.

A third key aspect of any species trait space relates to its structure and particularly how species are distributed and clustered in that space. Species with very similar traits are likely to play comparable roles in ecosystems (Dehling et al., 2016; Pigot et al., 2020; Sosiak & Barden, 2021) and are packed within a trait space into clusters (Figure 1d). The size (i.e., species richness) of these clusters relates to functional redundancy (Fonseca & Ganade, 2001; Walker, 1992), which could act as an insurance against the loss of certain combinations of traits and the disruption of ecosystem functioning under disturbance (McLean et al., 2019; Sanders et al., 2018). The other side of the same coin is functional uniqueness represented by species having no neighbours in the trait space owing to their unique combinations of traits (species B and D in Figure 1d). Several studies suggested that, beyond the positive influence of species trait diversity on ecosystem functioning (Craven et al., 2018; Gross et al., 2017), these unique species can play key and irreplaceable functional roles (Le Bagousse-Pinguet et al., 2021; Maire et al., 2018; O'Gorman et al., 2011; Pigot et al., 2016a). The filling of this trait space through evolutionary history, and more particularly the emergence of

species with unique traits, has also motivated numerous studies investigating specialisation in clades or competition footprint across the tree of life (Cornwell et al., 2014; Cox et al., 2021; Jarzyna et al., 2020; Phillips et al., 2018; Ricklefs, 2010; Stubbs & Benton, 2016). Yet we still lack a flexible framework in which the number and composition of species clusters but also unique species are automatically detected regardless of the shape, the density in terms of species richness and the dimensionality of the trait space in which they are embedded.

Here, we propose a unified and flexible framework to assess (1) the optimal number of axes representing species trait diversity (dimensionality), (2) the consistency of the trait space in species placement when sub-setting a limited number of traits (robustness) and (3) the distribution of species among clusters including the proportion of unique species (structure). To better understand the drivers of these three key aspects, we apply our framework on 30 trait datasets spreading across most kingdoms of life (e.g., bacteria, plants and vertebrates) and biomes (terrestrial and marine) at different scales (local to global), and spanning 2 orders of magnitude in species richness and 1 order of magnitude in the number of traits with different types (e.g., continuous and categorical) and varying proportions of missing values (Table 1). To disentangle the drivers of trait space complexity, we then model the dimensionality, the robustness and the structure of these 30 trait spaces as a function of the type of species, the type of ecosystem, the number of species, the number of traits, the type of traits, the correlation between traits and the proportion of missing values. Ultimately, we provide guidance to deal with the heterogeneity and incompleteness of species trait databases when building species trait spaces and assessing trait-based metrics in community ecology, evolution and biogeography.

# MATERIALS AND METHODS

## Building species trait space

Among the myriad of methods proposed to reduce the dimensionality of data (Kraemer et al., 2018; Laughlin, 2014; Nguyen & Holmes, 2019), we chose one that is commonly used in ecology, based on well-established ordination techniques, and flexible enough to be adapted to any kind of trait data. Our goal is not to review or compare existing methods but rather to assemble a suite of methods able to extract the main features of any species trait space and test their drivers.

First, we calculated trait dissimilarity between species pairs using the Gower pairwise distance (Gower & Legendre, 1986). This metric can handle multiple types of data (e.g., categorical, ordinal and continuous traits) and is also less sensitive to missing values than other distance estimation methods (Pavoine et al., 2009; Podani & Schmera, 2006). The dissimilarity between two species is only evaluated on traits with known values for both species, but this dissimilarity is standardised across all pairs whatever the number of traits considered. This step (Figure 1b) was carried out with the *daisy*() function in the *cluster* R package.

Second, we performed ordination of species in a space of reduced dimensionality by mean of principal coordinates analysis (PCoA), which identifies orthogonal axes along which trait dissimilarity is decomposed (Legendre & Legendre, 1998). For this step (Figure 1c), we used the *pcoa*() function in the *ape* R package.

## Quality of species trait space

To assess the dimensionality and robustness of species trait spaces, we needed a metric measuring the degree of distortion between the initial trait distance matrix between species pairs (Gower distance on all traits) and the distance matrix after dimensionality reduction (Euclidean distance on PCoA axes) or after removing traits (Gower distance on the sub-selection of traits), respectively. We assumed that a trait space is a high-quality representation of the full dataset if distances between species in that space are close to the initial distances computed with all traits (Maire et al., 2015). The approach of comparing the similarity of two distance matrices has precedent in Mantel tests (Legendre & Legendre, 1998), although the end goal here is quite different—producing a metric of robustness for low-dimensional trait space. Indeed, Mantel tests only correlate values or ranks between two distance matrices, ignoring the global co-ranking between species and their neighbourhood, which are key features of species trait space when the ultimate goal is to cluster species and identify functionally unique ones (Pimiento et al., 2020a).

Several measures of trait space quality have been proposed (Maire et al., 2015; Mérigot et al., 2010), but we chose a new one in the field of ecology with five key properties that overcome classical limitations: (1) being unitless so independent of the number, range or value of traits; (2) being standardised between 0 and 1 with a clear and intuitive interpretation of these extreme values; (3) avoiding the dilemma of whether or not to square the error, which arises in distance-based quality metrics; (4) being asymmetric by construction so only considering that the lower dimensional distance matrix is a poorer representation of species distribution in trait space compared with the initial distance matrix; and (5) proposing a common, albeit arbitrary, threshold to define quality.

This method is based on the co-ranking matrix $Q$ that compares the ranking of distance between objects in the initial distance matrix and in a lower dimensional space (Lee & Verleysen, 2009). In our case, let us denote by $\delta_{i,j}$ the distance between species $i$ and $j$ in the initial trait

matrix (Figure 1a) and $d_{i,j}$ their distance in the lower dimensional matrix (Figure 1c). Then, for any fixed species $i$, we assessed the ranks of the distances between this species $i$ and all other $S$-1 species $j$ in both the initial and lower dimensional matrices denoted as $\rho_{i,j}$ and $r_{i,j}$, respectively. These ranks varied between 1 and ($S$-1) with $S$ being the total number of species. The co-ranking matrix $Q$ is of size ($S$-1) by ($S$-1) and has for elements the number of species pairs that have the rank $k$ in the initial (all traits) Gower distance matrix and the rank $l$ in the lower dimensional (PCoA axes) Euclidean distance matrix (Figure 1e). Because the roles played by species $i$ and species $j$ are asymmetric, matrix $Q$ sums at $S(S\text{-}1)$, so the total number of pairs ($S$-1) made by each of the $S$ species.

Then we defined the rank error to be the difference $\rho_{i,j} - r_{i,j}$. If there is no error, that is, a perfect match in species neighbours between the initial and lower dimensional distance matrices, then $Q$ is a diagonal matrix; that is, ranks $k$ and $l$ will be similar, so $\rho_{i,j} - r_{i,j} = 0$ for all species pairs. At the opposite, rank mismatches or errors, due to dimensionality reduction or trait omission, induce off-diagonal species pairs in this co-ranking matrix (Figure 1e). These off-diagonal species pairs represent pairs that come at a lower distance rank (intrusion) or at a higher distance rank (extrusion) in the lower dimensional space compared with the initial space (Lee & Verleysen, 2009).

To assess whether the lower dimensional space was a good representation of the initial space, we needed an asymmetric measure. In other words, a measure that compares the ranks of species pairs in the lower dimensional matrix to those of the initial matrix and not the way around. A Spearman rank correlation is symmetric (the correlation between A and B equals the correlation between B and A) because it compares the ranks without any primary structure like in Mantel tests. We thus chose the area under the curve (AUC) criterion, which is based on the Somer's $D$ statistic, as an asymmetric rank measure (Somers, 1962). AUC is unitless and varies between 0 and 1. A value of 1 represents the best case scenario where the ranking of species pairs would be perfectly preserved between the initial and lower dimensional distance matrices (Kraemer et al., 2018). A rule of thumb to interpret this metric is that above 0.7, dimensionality reduction can be considered as good or acceptable and above 0.8 as excellent. Below 0.5, the lower dimensional space is a poor representation of the initial trait space whereas 0 means as good as random. It corresponds to the null or independence hypothesis in Mantel tests (Legendre & Legendre, 1998). More details can be found in Kraemer et al., (2018) who developed the *dimRed* and *coRanking* R packages for computing the co-ranking matrix $Q$ with the function *coranking* and then the AUC metric with the function *AUC_lnK_R_NX*.

Complementary to the AUC metric, which is only based on ranks so potentially weakly influenced by some extreme distortion values, we also compared the initial and lower dimensional distances between species pairs by using the Euclidean distance for multidimensional spaces, also known as the mean absolute deviation (MAD) (Maire et al., 2015).

## Dimensionality of species trait space

To determine how many dimensions are needed to build a trait space of enough quality that correctly positions species between each other, we used two approaches: a parsimonious one based on the elbow inflection point for the AUC metric and the other one based on a quality threshold for the AUC metric, both tested on 1 to 20 PCoA axes. The idea behind the elbow method is to maximise a given benefit (AUC gain in our case) while reducing the cost (number of dimensions in our case) (Thorndike, 1953). Consequently, the inflection point corresponds to the additional PCoA axis above which the benefit becomes lower than the cost (Figure S2). This elbow method is classically used in dimensionality analyses (Nguyen & Holmes, 2019) but never in combination with AUC.

As a complementary method, we used the AUC quality threshold of 0.7 to determine the dimensionality of the trait space so here the cumulated number of PCoA axes needed to obtain a good or acceptable positioning of species in the lower dimensional space compared with the initial one based on all traits. This approach is more subjective than the elbow one since it is based on an arbitrary threshold. However, it has the merit of providing a standardised, so comparable, quality value across datasets for the low-dimensional representations.

The amount of variance explained by the PCoA axes could also be considered as a quality metric of species trait space (Pimiento et al., 2020b) like with principal components analyses (PCA) (Pigot et al., 2020; Rüger et al., 2020). Yet, for non-Euclidean distances like Gower, PCoA axes may obtain negative eigenvalues corresponding to imaginary dimensions (Legendre & Legendre, 1998). In that case, the sum of all positive eigenvalues (real axes) is higher than the total variance of data. This intuitive additional piece of information was nonetheless included in our study through the examination of the relationship between the AUC-based dimensionality and the number of axes necessary to explain 50% of trait variation. The proportion of explained variance by PCoA axes was extracted using the ape::pcoa() *R* function.

## Robustness to trait omission

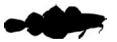
To test the robustness, or the lack of sensitivity, of the trait space to trait omission or sub-selection, we randomly removed between 10% and 80% (increments of 10%) of the total number of traits and then estimated a new Gower distance between all species pairs for each

**TABLE 1** We compiled 30 published trait datasets including 17 assembled in the CESTES database (Jeliazkov et al. 2020)

| Datasets | Taxon | Area | S | T | %NA | %Q | $\mu$ Cor | Icons |
|---|---|---|---|---|---|---|---|---|
| Biolog | Bacteria | Global | 865 | 97 | 0 | 99 | 0.17 |  |
| Bartonova et al. (2016) | Butterfly | Czech Republic | 128 | 13 | 0 | 100 | 0.22 |  |
| BirdLife | Bird | Global | 9297 | 20 | 0 | 100 | 0.12 |  |
| Carvalho et al. (2015) | Stream fishes | Amazonia, Brazil | 65 | 26 | 0 | 4 | 0.17 |  |
| Charbonnier et al. (2016) | Bird | Europe | 73 | 10 | 0 | 40 | 0.15 |  |
| Chmura et al. (2016) | Plant | Poland | 46 | 17 | 0 | 94 | 0.18 |  |
| Fish Base | Chondrichthyes | Global | 969 | 14 | 23 | 79 | 0.21 |  |
| Cleary et al. (2016) | Vertebrate | Jakarta Bay, Indonesia | 165 | 15 | 0 | 87 | 0.43 |  |
| Coral trait database | Invertebrate | Global | 802 | 12 | 25 | 42 | 0.12 |  |
| Mellado-Diaz et al. (2008) | Invertebrate | Segura River, Spain | 208 | 62 | 0 | 0 | 0.12 |  |
| Eallonardo et al. (2013) | Plant | New York State, USA | 41 | 11 | 0 | 55 | 0.24 |  |
| Toussaint et al. (2016) | Freshwater fish | Global | 8134 | 10 | 3 | 100 | 0.10 |  |
| Fried et al. (2012) | Plant | France | 75 | 10 | 0 | 30 | 0.17 |  |
| Gibb et al. (2015) | Spider | South-Eastern Australia | 86 | 10 | 0 | 100 | 0.41 |  |
| Gonçalves-Souza et al. (2014) | Spider | Brazilian coast | 112 | 21 | 0 | 95 | 0.32 |  |
| Jeliazkov (2013) | Macroinvertebrate | France | 112 | 89 | 0 | 0 | 0.14 |  |

(Continues)

**TABLE 1** (Continued)

| Datasets | Taxon | Area | S | T | %NA | %Q | $\mu$ Cor | Icons |
|---|---|---|---|---|---|---|---|---|
| Krasnov et al. (2015) | Ectoparasite | Palearctic | 177 | 12 | 0 | 100 | 0.17 |  |
| Loiseau et al., (2020) | Terrestrial mammals | Global | 4675 | 15 | 0 | 73 | 0.14 |  |
| McLean et al., (2018) | Fish | North Sea, Atlantic | 138 | 14 | 3 | 64 | 0.13 |  |
| Doledec et al. (2011) | Stream macroinvertebrate | New Zealand | 495 | 59 | 0 | 0 | 0.14 |  |
| Pakeman (2011) | Plant | Scotland | 148 | 28 | 0 | 36 | 0.14 |  |
| Kissling et al. (2019) | Plant | Global | 2557 | 22 | 28 | 82 | 0.17 |  |
| Pavoine et al. (2011) | Plant | Algeria | 56 | 14 | 0 | 29 | 0.15 |  |
| Rimet & Druart (2018) | Phytoplankton | Temperate lakes | 1222 | 15 | 0 | 40 | 0.30 |  |
| Thuiller et al., (2014) | Plant | French Alps | 3718 | 33 | 16 | 12 | 0.12 |  |
| Ribera et al. (2001) | Beetle | Scotland | 68 | 20 | 0 | 50 | 0.17 |  |
| Chapman et al. (2019) | Thermal vent | Global | 646 | 16 | 15 | 31 | 0.18 |  |
| USDA (2020) | Plant | USA | 1876 | 20 | 6 | 90 | 0.09 |  |
| Villéger et al. (2012) | Fish | Mexico | 46 | 16 | 0 | 100 | 0.19 |  |
| Yates et al. (2014) | Ant | New South Wales, Australia | 123 | 11 | 0 | 91 | 0.18 |  |

*Note:* We selected datasets with at least 40 species and 10 traits for high-dimensional space and perform sub-selection analyses (Figure S1). Further, to allow the calculation of trait dissimilarity between all species pairs and test of robustness to trait omission, we removed traits with missing information for more than 60% of species and species with more than 50% of missing information for their traits. This procedure altered 4 out of 30 datasets and on average removed 15% (7%–25%) of species and 57% (22%–79%) of traits per dataset. We provide a description and a reference of each dataset with the geographic extent and location (Area), species richness (S), the number of traits (T), the percentage of missing values (%NA), the percentage of quantitative traits (%Q) (the others, 1–%Q, being categorical), the mean pairwise Kendall correlation between traits (Cor) and a unique icon for each taxon used in some other figures.

removal percentage; we did not use PCoA axes in this robustness analysis, only traits. Then we assessed the level of congruence between the initial distance matrix and the lower dimensional distance matrix by computing the AUC and MAD metrics. These simulations were performed 100 times for each removal percentage. We then extracted an index of robustness defined as the opposite of sensitivity so the mean loss of AUC when 50% of the traits are removed.

## Species clustering and uniqueness

To cluster species in the trait space and potentially identify unique species, we used the 'clustering by fast search and find of density peaks' algorithm, which is based on initial pairwise distances and does not require dimensionality reduction (Rodriguez & Laio, 2014). Yet the robustness of the clustering critically depends on the robustness of pairwise species distances to trait omission. Among the many clustering algorithms that have been proposed (Condon et al., 2016; Jain & Dubes, 1988; Xu & Tian, 2015), this one combines the advantages of (1) clustering objects regardless of the shape and dimensionality of the space in which they are embedded, (2) detecting isolated objects automatically independently of their number and (3) making the number and size of clusters emerge with no a priori expectation or arbitrary choice.

In our case, this algorithm first computed the density of neighbours for each species, defined as the number of species that are within a given small distance $d_0$ (Figure 1f). Given this density, the algorithm then relied on two basic principles: (1) cluster centres were species characterised by a higher density of neighbours than their own neighbours and by a relatively large distance from other species with a higher density of neighbours, and (2) isolated or unique species had no neighbours at maximum $d_0$ (zero density or redundancy). Once cluster centres and unique species were identified, all remaining species were assigned to a cluster corresponding to the nearest neighbour of higher density (Rodriguez & Laio, 2014). We adopted two modifications to reduce arbitrary choices. First, the identification of cluster centres was fully automated: all species with higher neighbour density than their own neighbours and at a distance of at least $d_0$ from species with higher density were considered as cluster centres. Second, if two clusters were not separated by a 'low density valley', that is, a region of radius $d_0$ where densities were lower than those of the cluster centres, they were merged.

The whole clustering process thus required only a single free parameter, the threshold $d_0$, fixed by a rule of thumb by which the minimum distance to the nearest neighbour defining isolation, that is, species uniqueness in trait space, is the average number of neighbours around each object corresponding to 1% or 2% of the total number of species in the dataset (Rodriguez & Laio, 2014). This procedure has the advantage of not fixing a $d_0$ value a priori for all datasets but instead to define a $d_0$ value for each dataset only depending on species number. Unique species can thus be considered as relative isolates in the trait space. We chose 1% as a conservative rule to not cluster species being too different in traits so keeping $d_0$ small. We provide an R implementation of this algorithm along with the code to reproduce all the analyses of this paper (R Core Team, 2021; see section Data and Code availability).

## Influence of trait dataset characteristics

To test whether the characteristics of species, ecosystems and traits can influence the dimensionality, robustness and structure of species trait space, we performed general linear models (GLMs) with a Gaussian distribution for all response variables, that is, the elbow-based dimensionality, the threshold-based dimensionality, the robustness to 50% trait removal, the log-transformed number of species clusters, the percentage of species packed in the first cluster and the percentage of unique species (distributions are shown in Figure S3). As explanatory factors, we used the type of species life form (plant, invertebrate and vertebrate) and the type of ecosystem (aquatic and terrestrial) to test the potential effects of organismal complexity. We also used the log-transformed number of species and number of traits as the dimensions of the initial species trait matrix. Trait characteristics were then used as potential drivers like the percentage of missing values, the percentage of quantitative traits and the mean pairwise correlation between traits, expressed as the rank-based Kendall index able to mix continuous and categorical traits. Pairwise correlations between quantitative trait dataset characteristics are rather low ($-0.19 < r < 0.45$) and mainly non-significant (Figure S4).

We then used partial regression plots to highlight the effect of each factor while controlling for the others (set at their mean). Statistical analyses were carried out using the function *glm* from the *stats* R package whereas partial plots were drawn using the function *visreg* from the *visreg* R package.

In addition to the analyses performed on empirical datasets, we also built three simulated datasets to test the effect of species and trait number on the dimensionality of species trait space without changing the type of traits as a controlled experiment. Continuous traits for 1000 species were generated following a uniform distribution (0–1) with no missing value. In the first dataset, we simulated 10 uncorrelated traits, in the second 10 correlated traits ($r = 0.5$) and in the third 20 uncorrelated traits. We then estimated the trait space dimensionality for each level of species number and each dataset using the AUC threshold of 0.7.

# RESULTS

## Trait space dimensionality

Over the 30 datasets, we obtained an optimal reduced dimensionality ranging between 2 and 8 axes (median =4) using the elbow method and between 2 and 17 axes (median = 6) using the AUC threshold of 0.7 when attained. For all datasets, we could reach the AUC threshold of 0.7 with less than 20 dimensions or PCoA axes, except for plants of the French Alps for which AUC remained low (<0.6) even with many axes (Figure 2). For the remaining 29 datasets, the correlation between the elbow-based and threshold-based dimensionality was positive but weak ($r$ = 0.3) and non-significant ($p$-value =0.10), highlighting their complementarity (Figure S5). With a more demanding threshold of AUC =0.8 (high-quality trait space), up to 24 datasets could reach this value with a maximum of 20 dimensions (Figure 2).

Two first GLMs, including all explanatory factors but only 29 datasets out of 30 (bacteria were excluded because they are the only representative of a kingdom), showed that the type of life form (plant, invertebrate and vertebrate) and the type of ecosystem (aquatic and terrestrial) did not significantly explain the elbow-based and threshold-based dimensionality (Table S1). The partial regression plots illustrate these weak influences while controlling for the other factors (Figure 3). We thus retained only quantitative variables related to the characteristics of the species trait datasets in the following analyses.

The elbow-based dimensionality was weakly explained by the five quantitative characteristics of the datasets ($R^2$ = 0.15), but the correlation between traits had by far the main effect, albeit non-significant ($p$-value =0.09) (Table S2), with a lower optimal number of axes when the correlation between traits increased (Figure 4). The threshold-based dimensionality was well explained by characteristics of the datasets ($R^2$ = 0.61) with the log number of traits and the correlation between traits having the strongest and only significant effects (Table S2). The partial regression plots showed that the threshold-based dimensionality strongly increased with the log number of traits whereas it decreased with the

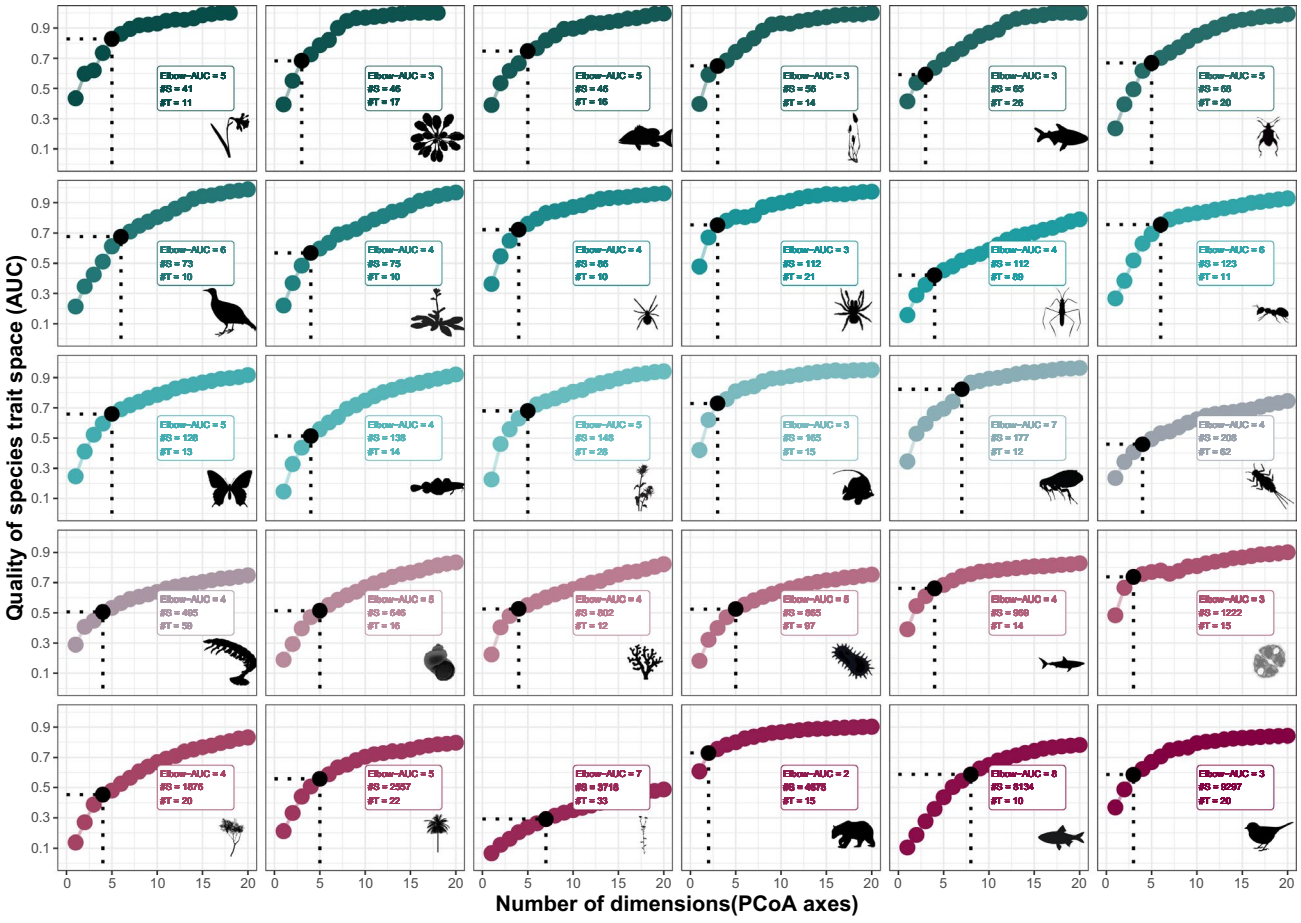



**FIGURE 2** Influence of the number of dimensions (number of retained PCoA axes) used to build the 30 species trait spaces on the space quality assessed by the area under the curve (AUC) criteria. The black dots and dotted lines correspond to the elbow-based optimal dimensionality for each dataset. The values indicate the elbow-based dimensionality, the total species richness (#*S*) and the total number of traits (#*T*) in each dataset. Datasets are ranked (top left to bottom right and from dark green to dark red) following the number of species

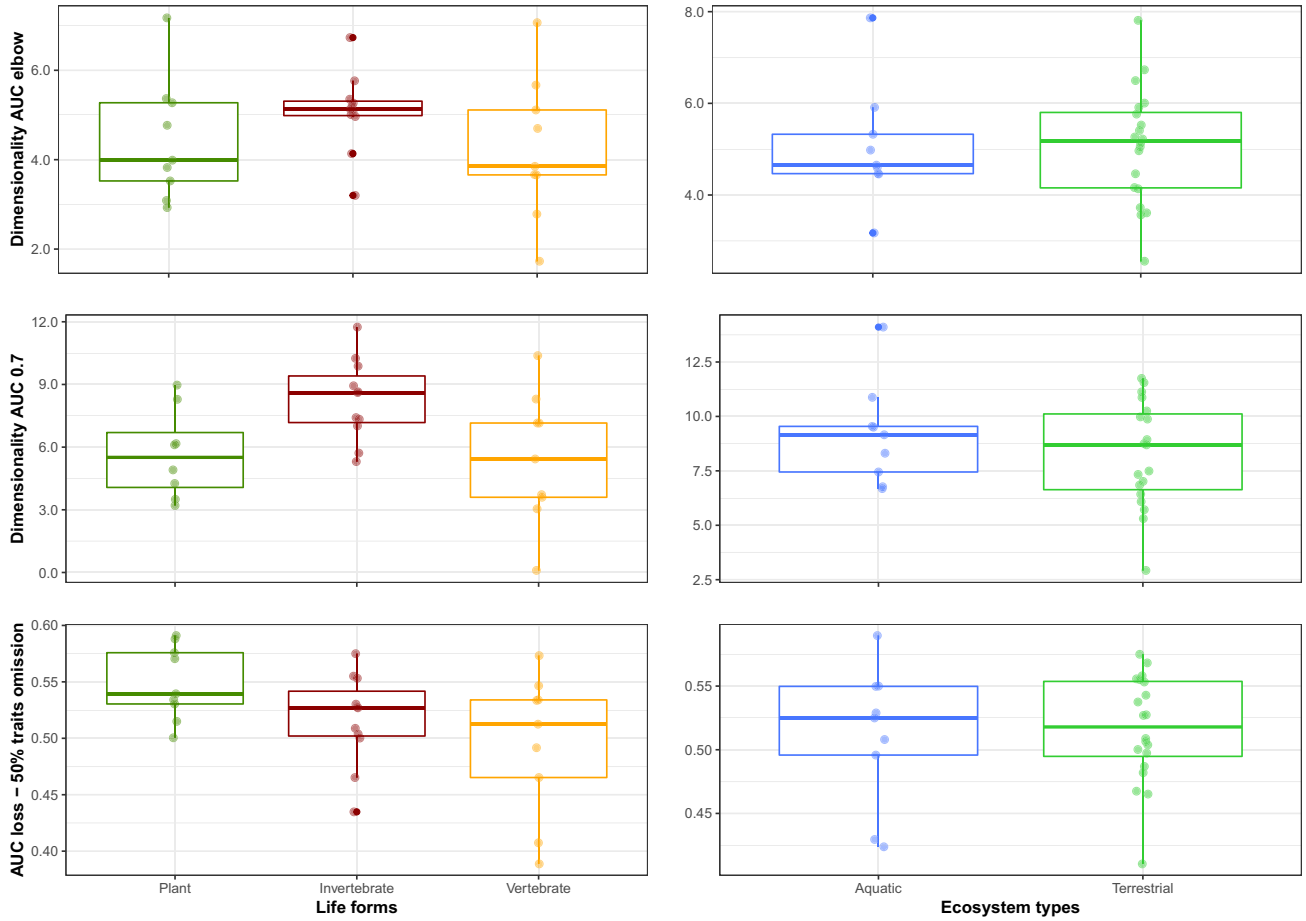



**FIGURE 3** Partial plots showing the influence of the species life form (plant, invertebrate and vertebrate) and ecosystem type (aquatic and terrestrial), while controlling for the five dataset quantitative characteristics, on species trait space dimensionality measured with the elbow-based (first row) or threshold-based (second row) AUC criteria. The third row shows trait space robustness, in terms of AUC loss, to trait removal or omission (50%) according to the two factors being tested. Related statistics are reported in Table S1; the effects are all non-significant

correlation between traits (Figure 4). As a complementary analysis, our simulated trait datasets confirmed the main influence of the number and the correlation of traits on species trait space dimensionality whereas the number of species had only an effect for less than 100 species and no effect above 200 species (Figure S6).

The number of axes necessary to explain 50% of trait variation was a weak predictor of the elbow-based dimensionality ($R^2 = 0.18$) but was a strong predictor of the threshold-based dimensionality ($R^2 = 0.82$), albeit underestimated (Figure S7).

## Robustness to trait omission

The robustness to trait omission was generally low over the 30 datasets with a mean AUC loss of 0.54 ($SD$ =0.12) when 50% of the traits were deleted. In these cases, most low-dimensional trait spaces were poor representations of the initial distances between species. Yet this robustness was highly heterogeneous among datasets ranging from 0.33 to 0.85 of AUC loss (Figure 5). To stay above the AUC threshold of 0.7, trait omission should not exceed 20% on average when we ignored the five datasets for which even removing 10% of traits induced an AUC loss of more than 0.3 (i.e., AUC <0.7).

Like for the dimensionality, the robustness to trait omission was not significantly influenced by either the type of species life form or the type of ecosystem (Figure 3 and Table S1), so these factors were ignored in the following analyses focused on quantitative factors. The robustness to trait omission was strongly dependent on the dataset characteristics ($R^2$ = 0.84) with the log number of traits, the percentage of missing values and the correlation between traits having the strongest and only significant effects (Table S2). The partial regression plots revealed quite logically that the robustness to trait omission (opposite to AUC loss) increased with the number of traits but also with the correlation between traits (Figure 4). In contrast, robustness was negatively related to the percentage of missing values, which again makes sense. With many missing values, the trait space is likely to be unstable under trait omission, so dimensionality reduction may distort the representation of the initial distances between species.

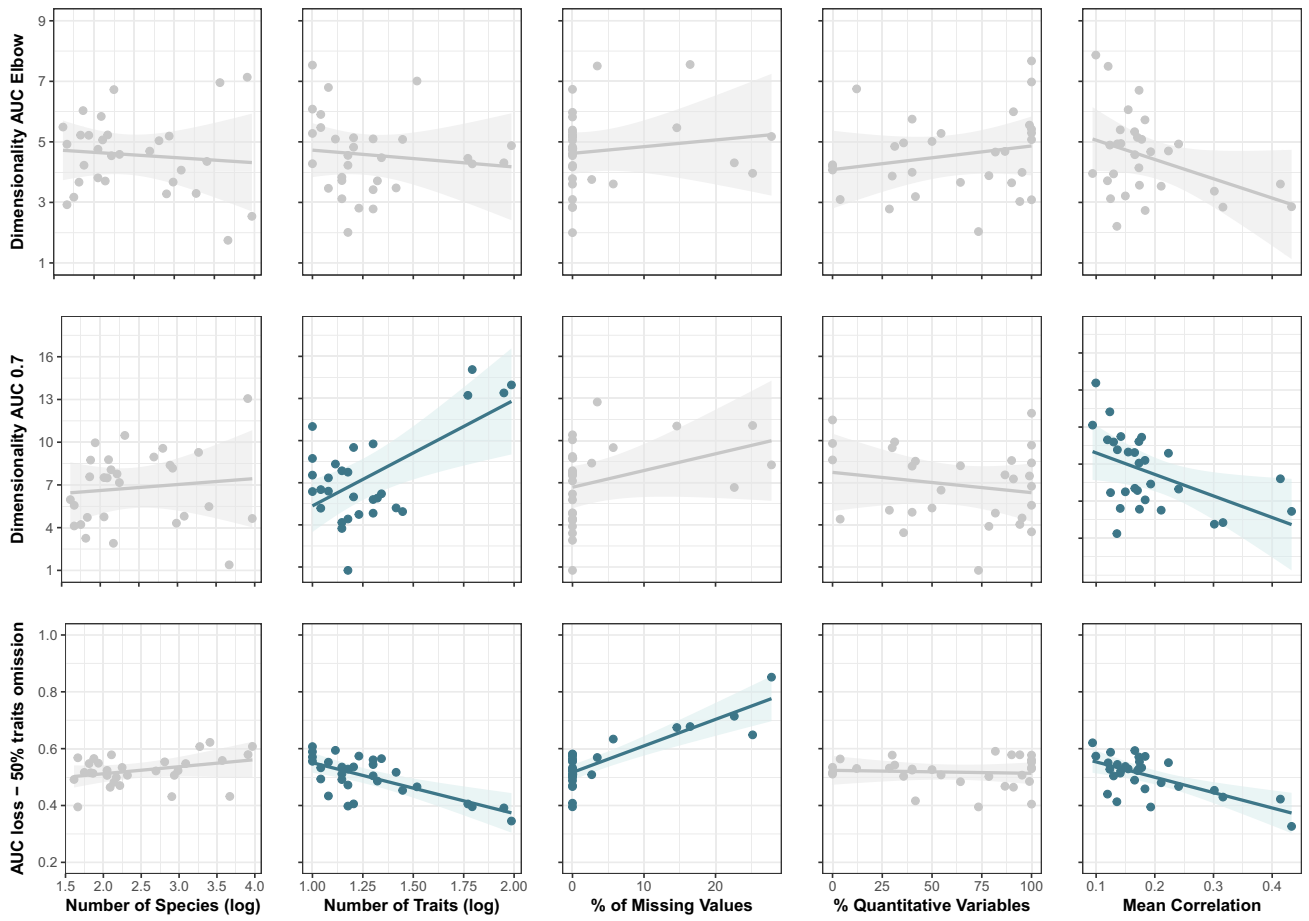



**FIGURE 4** Partial plots showing the influence of the five trait dataset characteristics on species trait space dimensionality measured with the elbow-based (first row) or threshold-based (second row) AUC criteria. The third row shows trait space robustness, in terms of AUC loss, to trait removal or omission (50%) according to the five characteristics. Only significant ($p < 0.05$) relationships are coloured; the others are grey. Related statistics are reported in Table S2

## Species clustering in trait space

Over the 30 datasets, the number of species clusters, delineated by the 'fast search and find of density peaks' algorithm, varied between 4 and 434 and was moderately explained by the dataset characteristics ($R^2 = 0.57$). The number of clusters was not significantly influenced by either the type of species life form or the type of ecosystem (Figure 6 and Table S3), so these factors were ignored in the following analyses. The main and only significant drivers were the log number of species and percentage of missing values (Table S4). The number of clusters logically decreased with the percentage of missing values because less trait combinations can be realised but increased with the number of species (Figure 7 and S8). Yet the number of clusters increased as a saturating power law with the number of species owing to a slope much lower than 1 (0.41) in the log–log relationship when we controlled for other effects (Figure 8a).

The proportion of species belonging to the first or dominant cluster was not significantly driven by either the type of species life form or ecosystem (Figure 6 and Table S3), so these factors were ignored in the following analyses. This species packing into the dominant cluster was mainly driven by the log number of species with a predictive power of $R^2 = 0.58$ whereas all the other dataset characteristics had non-significant influences (Table S4). The slope of the relationship between the proportion of species clustered within the first group and the log number of species was positive (Figure 7), highlighting that species tended to pack in the richest trait cluster when species richness increased, regardless of the other dataset characteristics. Yet the log–log relationship between the total species richness and the richness of the first cluster revealed a power law with a slope higher than 1 (1.38) when we controlled for other effects (Figure 8b), suggesting that species packing disproportionately increased with species richness.

## Unique species in trait space

The number of unique species, that is, species that did not belong to any cluster so isolated in the trait space, varied between 27 and 1750 among datasets with a percentage ranging from 2% to 74% (median =42%). These unique

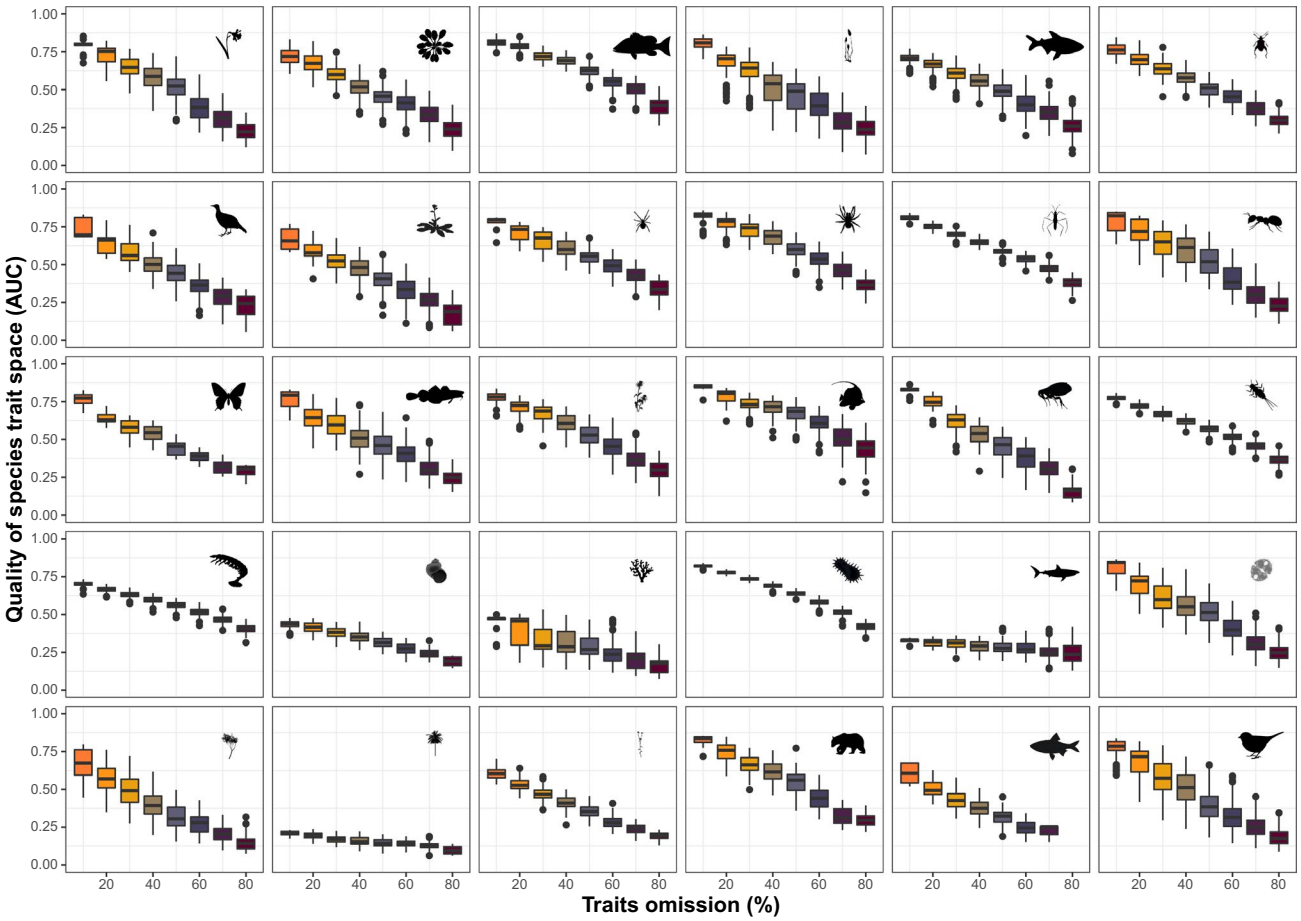



**FIGURE 5** Influence of the percentage of traits omission (between 10% and 80%) on the quality of the trait space in terms of AUC when representing species in a trait space of lower dimensionality. For this, we randomly removed traits 100 times for each level of omission to obtain the boxplots across the 30 datasets ranked by the total number of species (top left to bottom right). For 0% of trait omission, AUC is 1

species were widespread in trait space and not just located on the edges, suggesting openings scattered throughout species trait spaces (Figure 9). Yet well-known unique species appeared clearly far on the edge such as the whale shark (*Rhincodon typus*), which is the largest shark (20 m long and body mass of 34 tonnes) while being a planktivore, so an ecological outlier among Chondrichthyes.

The proportion of unique species was not significantly influenced by either the type of species life form or ecosystem (Figure 6 and Table S3), so these categorical factors were ignored in the following analyses only based on quantitative factors. The proportion of unique species was strongly explained by dataset characteristics ($R^2 = 0.82$) with the log number of species and, to a less extent, the percentage of missing values, being the main drivers (Table S4).

The partial regression plots revealed that the proportion of unique species had a marked negative relationship with the log number of species while controlling for other effects (Figure 7), suggesting that species-rich assemblages left less space for ecological uniqueness or that species tended to disproportionately pack into the richest cluster when diversity increased (Figure 8b). This saturating relationship was highlighted by the partial plot linking the total number of species and the number of unique species with a power log–log slope of 0.47 (Figure 8c). The proportion of unique species also decreased with the proportion of missing values because it mechanically reduced the diversity of trait combinations and increased species similarity (Figure 7).

## DISCUSSION

### The necessary trade-off between trait space quality and operationality

Trait-based approaches have a long tradition in life science since the development of the two-strategy life-history framework from 'fast' (r) to 'slow' (K) organisms (MacArthur & Wilson, 1967; Pianka, 1972). This oversimplified view was later extended to triangular continuums of plant life-history strategies with the well-known competitive ability—physiological tolerance to stress—adaptation to disturbance (C-S-R) schema introduced by Grime (1977) and the leaf-height-seed (LHS) framework by Westoby (1998). Such meaningful simplifications

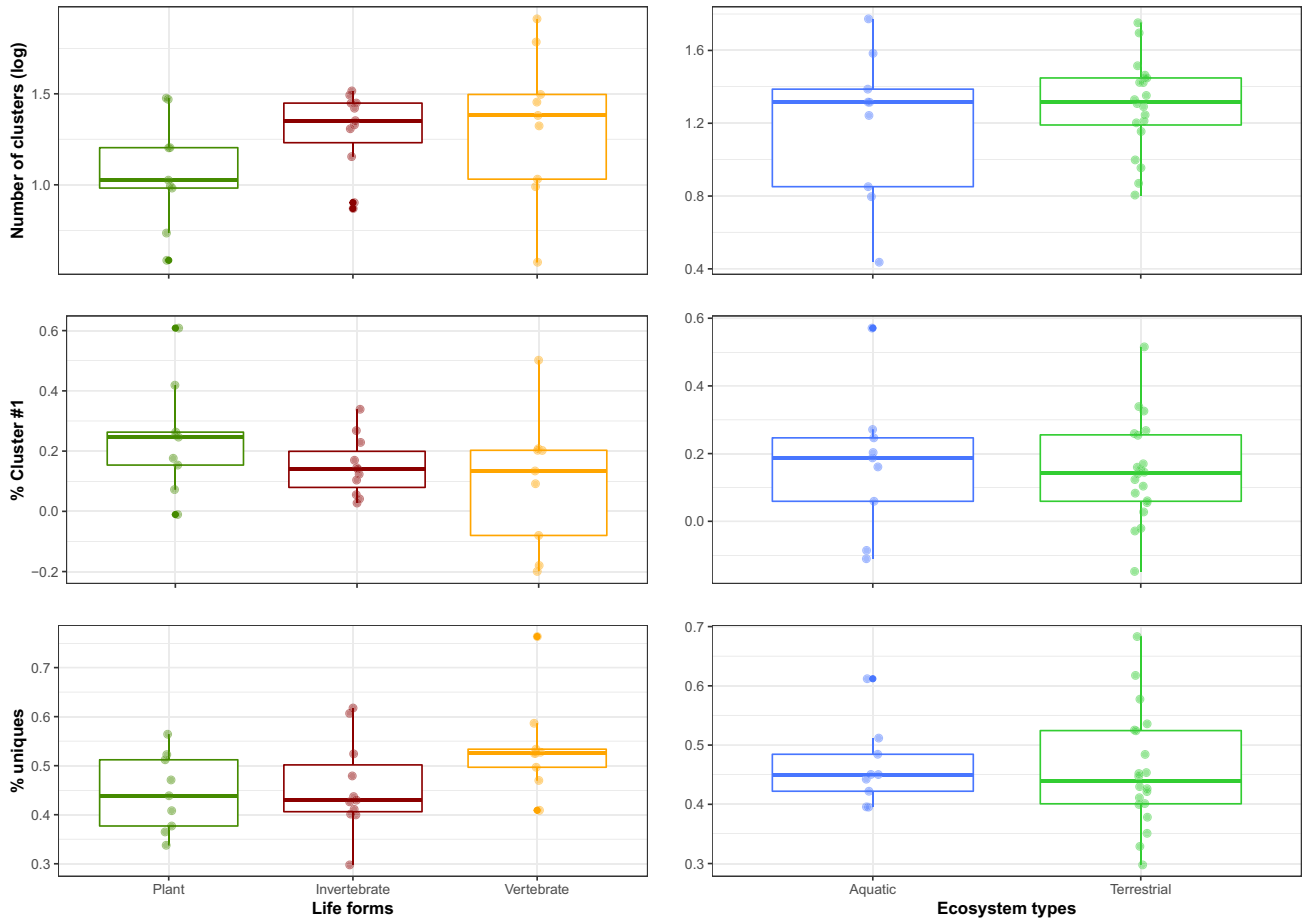



**FIGURE 6** Partial plots showing the influence of the species life form (plant, invertebrate and vertebrate) and ecosystem type (aquatic and terrestrial), while controlling for the five dataset quantitative characteristics, on the log number of species clusters (first row), the proportion of species packed in the first or dominant cluster (second row) and the proportion of unique species so those isolated in the trait space (third row). Related statistics are reported in Table S3; the effects are all non-significant

of trait variability among species have revolutionised functional ecology and inspired similar successful approaches for insects (Greenslade, 1983), freshwater fishes (Winemiller & Rose, 1992), corals (Darling et al., 2012) and microbes (Malik et al., 2020). In the case of well-established or experimentally tested causal relationships between traits and environments or functions, the dimensionality issue is of marginal importance when building species spaces with few relevant traits delineating clearly defined ecological strategies. By contrast, when such knowledge is lacking, so when many traits are available with low evidence of particular causal relevance, when big data analyses are performed with many missing values or when species strategies cannot be summarised by a limited set of traits, ecologists face the challenge of trait space hyper-dimensionality (Blonder et al., 2014).

Dimensionality reduction can then be a necessary step because some widely used functional diversity indices (e.g., functional richness) are based on the volume of trait space (convex hull volume) occupied by species of a given ecosystem (Laliberte & Legendre, 2010; Trindade-Santos et al., 2020; Villeger et al., 2008) that can be hardly calculated beyond six dimensions, even less (four to five) if null models are required or when pairwise site measures like $\beta$-diversity have to be estimated (Loiseau et al., 2017; Pimiento et al., 2020b; Su et al., 2021; Villeger et al., 2011). Because most common functional diversity indices are sensitive to the degree of correlation among traits (Zhu et al., 2017), we also suggest to compute these indices from a reduced number of independent PCoA axes to improve the capacity to distinguish between communities along gradients of stress (Trindade-Santos et al., 2020).

Beyond practical reasons, this dimensionality value also informs about the extent to which species traits can be reduced to a limited number of ecologically meaningful axes (Díaz et al., 2016; Pigot et al., 2020). This quest for ecological syndromes or strategies is not new (Reich et al., 2003; Westoby, 1998), and some previous studies have investigated the intrinsic dimensionality of species traits using various linear and non-linear methods (Laughlin, 2014; Maire et al., 2015; Westoby, 1998). Here, we proposed two complementary ways to estimate linear dimensionality, and we applied them to 30 datasets to ultimately identify their main drivers, if any.

Using the parsimonious elbow-based AUC method, we found a median dimensionality of 4 axes, which is a

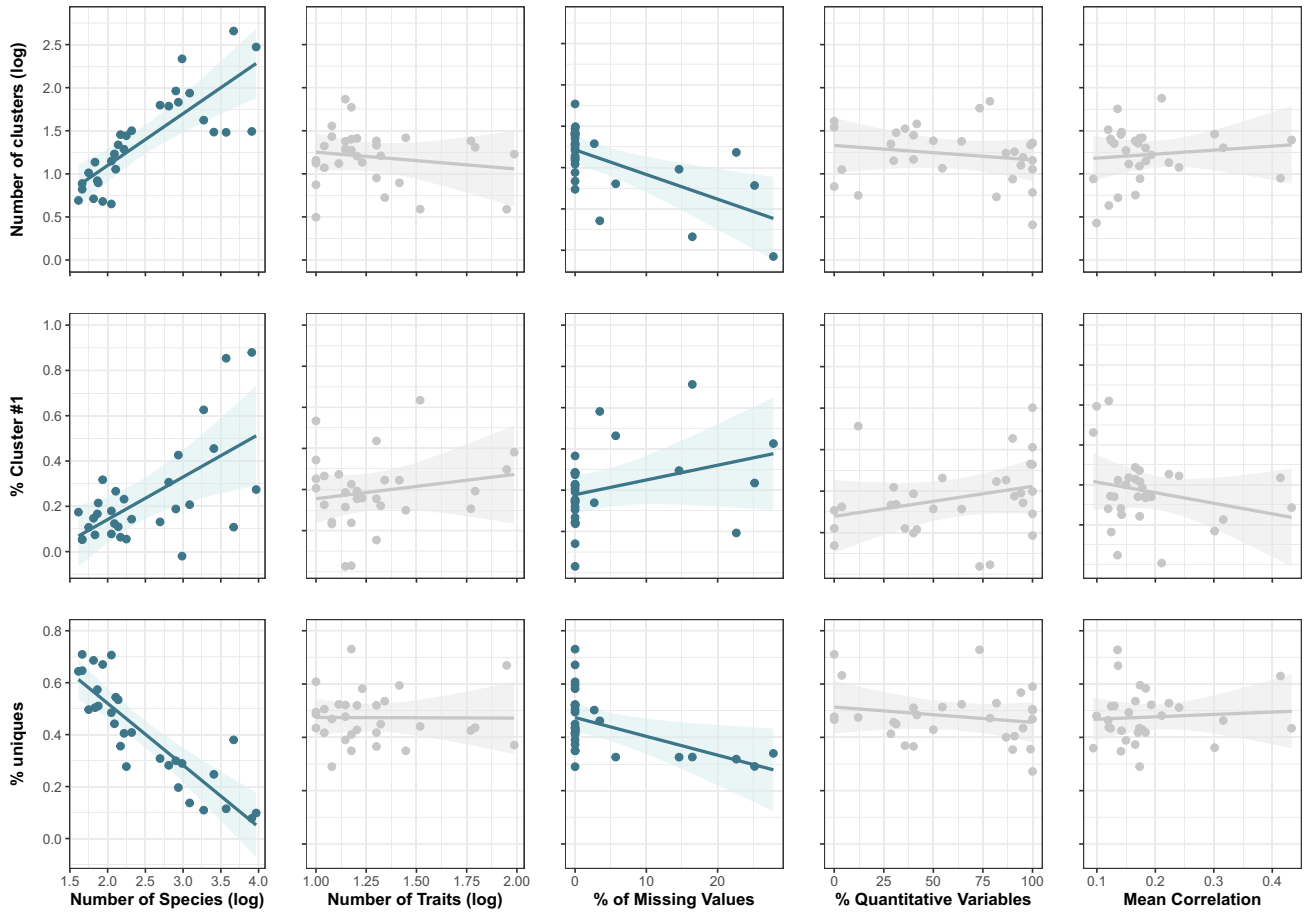



**FIGURE 7** Partial plots showing the influence of the five trait dataset characteristics on the log number of species clusters (first row), the proportion of species packed in the first or dominant cluster (second row) and the proportion of unique species so those isolated in the trait space (third row). Only significant ($p < 0.05$) partial relationships are blue plain dots and lines; others are in grey. Related statistics are reported in Table S4

rather low value given that we only considered datasets with at least 10 traits in our study (Table 1). Interestingly, for most datasets (25 out of 30), the elbow-based dimensionality is lower than 6 axes (2–5) (Figure 2), suggesting that the calculation of most volume-based functional diversity indices can be performed even with null models. Using the AUC threshold criteria of 0.7, the dimensionality is higher (median of six axes) and generally out of the operational range for calculations of hypervolume-based metrics like functional richness (Villeger et al., 2008) or functional $\beta$-diversity (Loiseau et al., 2017). It reinforces the idea that the diversity of organism forms and functions has a larger dimensionality than previously thought (Messier et al., 2017; Pigot et al., 2016b) whatever the kingdom and ecosystem. Only poor assemblages (<30 species) can be accurately described with low dimensionality (<4 axes) as shown in our simulations (Figure S6).

This can be partly due to the coexistence of different syndromes related to different sets of traits, corresponding to different ecological strategies, under a given environment (Reich et al., 2003; Sosiak & Barden, 2021). For instance, landscape filters can shape trait community composition with species sharing some traits (trait syndromes) responding in a similar way under the same environmental conditions (e.g., agricultural intensification) (Gámez-Virués et al., 2015). When using large species datasets mixing various environments and many traits like in most of our cases (Table 1), the potential multiplication of trait syndromes could explain the relatively high dimensionality in the trait space we have observed, particularly for the plants in the French Alps or stream macroinvertebrates (Figure 2). We may expect lower dimensionality in species trait space built from local communities under severe filters owing to the predominance of a few but highly constrained trait syndromes. We may also expect lower dimensionality when using effect versus response traits in a more coherent and systematic manner with a clear defined goal (Luck et al., 2012).

The most surprising result is the weak positive correlation between the elbow-based and threshold-based dimensionality values showing that a low elbow-based AUC value does not imply passing the 0.7 AUC threshold and vice versa (Figure 2). This is because the elbow-based method imposes a compromise between the quantity of axes and the quality of the trait space

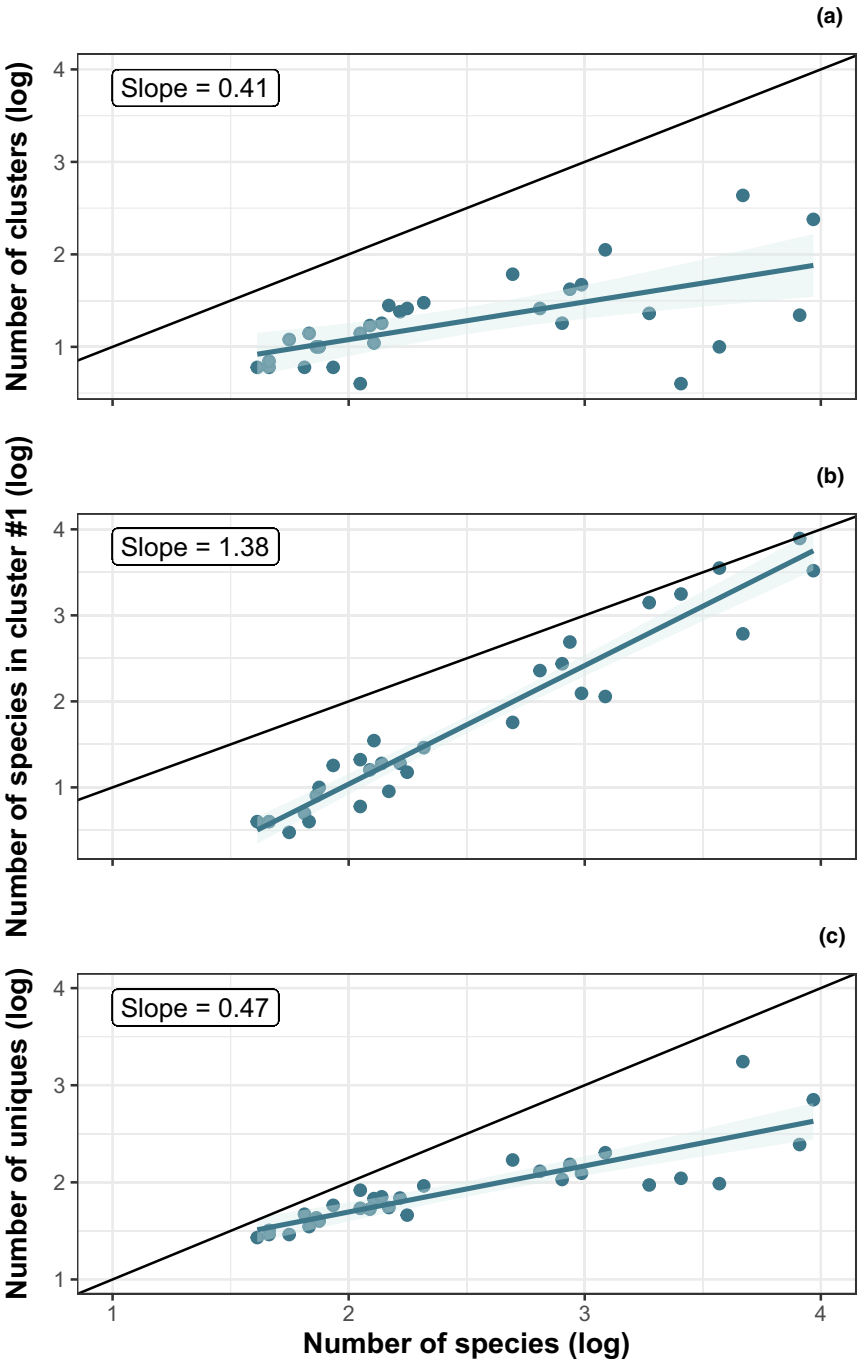



**FIGURE 8** Partial log–log relationships between the number of groups clustered by the fast search and find of density peaks algorithm (a), the number of species in the most dominant cluster (b) and the total number of unique species so those not being part of any group (c), and the number of species in the 30 datasets. Slopes of the log–log relationships, so exponents of power laws, are reported

to avoid selecting more poorly informative axes (over-dimensionality) whereas the threshold-based method only considers quality whatever the quantity of axes. Given this constraint, the elbow-based method provides lower dimensionality values (2–8 axes against 2–17 axes for the threshold method; Figure S5), which are also less influenced by dataset characteristics. As a practical guide, we suggest to use the elbow-based method as a first estimate of dimensionality on a given trait dataset and then to increase the number of dimensions to be considered until passing the 0.7 threshold if necessary. With this rule of thumb, we should end up with an optimal dimensionality comprising between 3 and 6 axes for most datasets, as a trade-off between operationality and quality. Obviously, the operational constraint depends on species number, diversity indices being used and power facilities.

In case a value of AUC =0.5 cannot be reached with a reasonable number of dimensions (<10 axes) like on the French Alps plants (Figure 2), we suggest either to

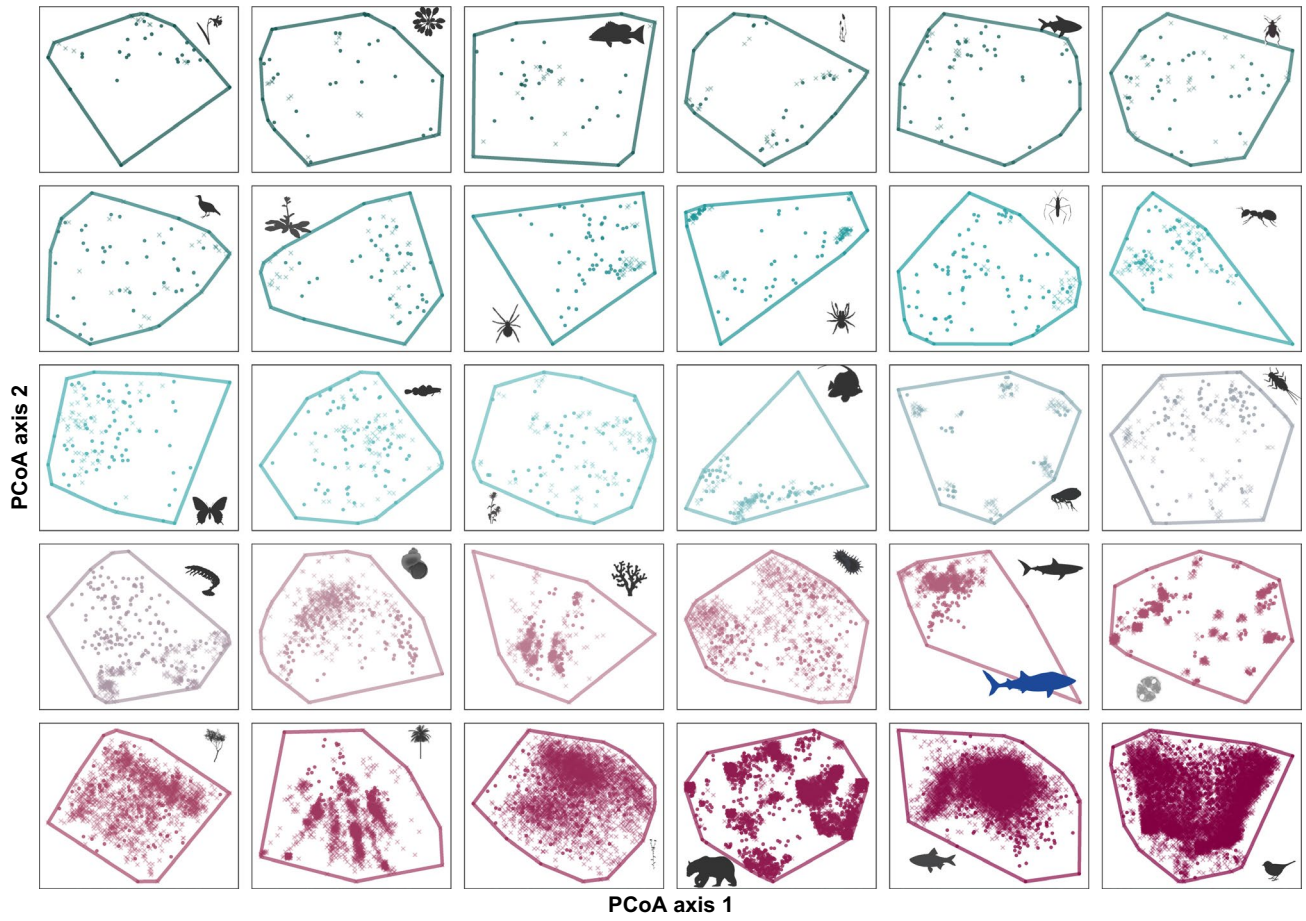



**FIGURE 9** Trait spaces for the 30 datasets where the two axes come from principal coordinates analyses (PCoA) representing the distribution of species according to their trait values. Species coloured in dark are detected as statistically and ecologically unique species by the fast search and find of density peaks algorithm. The whale shark (*Rhincodon typus*) is highlighted in blue being highly distinct and unique in its clade. Datasets are ranked (top left to bottom right and from dark green to dark red) following the number of species

carefully select the most relevant traits given the question being addressed (Thuiller et al., 2014) or to avoid indices based on trait space reduction (like functional richness) but instead to use distance-based indices (Rao) only (Chao et al., 2019; Laliberte & Legendre, 2010; Mouillot et al., 2013). For representation purposes, which are classically drawn in two or three dimensions with PCoA axes 1 to 4 (Bruelheide et al., 2018; Loiseau et al., 2020; Pimiento et al., 2020b; Stubbs & Benton, 2016), we suggest to provide the corresponding AUC value as a key information on trait space quality along with the percentage of trait variation explained by axes. Because dimensionality is weakly influenced by dataset characteristics, except trait correlations that decrease dimensionality for both elbow-based and threshold-based criteria (Figure 4), we suggest to pay particular attention to unnecessary or meaningless traits that are strongly independent from the others and would inflate dimensionality potentially biasing biodiversity metrics. Conversely, considering redundant or correlated traits, even if meaningless, has no expected impact on dimensionality so can be very neutral in the building of species trait space and the computation of indices. Yet using surrogate traits or traits with a coarse resolution to describe a given dimension of ecological strategy can substantially affect the results (Kohli & Jarzyna, 2021; Loranger et al., 2016).

## The low but predictable robustness to trait omissions or choices

Choosing a set of traits always means ignoring some, whereas important traits can be missed because they are unavailable or unknown. Often traits are ignored for non-biological reasons such as the difficulty of measuring them or the lack of standardisation in the research community. The consequences of this sub-selection have been poorly investigated, despite its potential to modify the perceived dissimilarity between species (Carscadden et al., 2017) and profoundly affect the estimates of functional diversity (Zhu et al., 2017). Here, we randomly reduced our trait datasets to assess the impact of trait omission on AUC loss between the initial distance matrix

(all traits) and that based on 90% to 20% of the traits only (Figure 5). When only 10% of traits are removed, AUC is still higher than 0.7 on average across simulations in 21 datasets out of 30, suggesting overall high robustness to low rate of trait omission except for some taxa like palm trees, sharks, thermal fauna and corals that belong to different kingdoms and ecosystems. At 50% of trait removal, AUC severely drops below the 0.7 threshold for all datasets except fishes of the Jakarta Bay (Figure 5).

This overall low but highly variable robustness of species distances to trait omission is very well explained by dataset characteristics (Figure 4). Unsurprisingly, AUC loss at 50% omission rate is negatively related to the number of traits, so that trait-poor datasets (corals, sharks or freshwater fishes) are more sensitive to the removal of traits than their trait-rich counterparts (macroinvertebrates or bacteria). Our statistical model also shows an expected negative relationship between AUC loss and trait correlation, so with more redundant traits, the distances between species pairs in a low-dimensional space are more strongly preserved. This might explain why dimensionality reduction has been successful for some research fields in functional ecology (e.g., leaf traits and the leaf economic spectrum, Díaz et al., 2016; Wright et al., 2004), whereas other studies such as those spanning many organs of plants have failed to find meaningful reduction in trait dimensionality (Carscadden et al., 2017; Messier et al., 2017). We also point out that the number of missing values strongly impacts robustness to trait omission, so including traits with many missing values (>10%) can be a counterproductive effort, especially with Gower-like metrics that only consider traits with no missing values to assess the distance between two species. We also show a high variability in robustness for a given level of trait omission (Figure 5), suggesting that robustness to trait omission depends on traits being removed, some being more critical than others, independently of their ecological relevance. This reinforces the advice to carefully select traits prior to analyses and pay a particular attention to those being uncorrelated to the others given their disproportionate importance in the structuring of species trait spaces and subsequent analyses.

Taken together, these results point out that the robustness of species space to trait omissions or choices is on average lower than previously thought (Douma et al., 2012) and that dataset characteristics, not the species life form or ecosystem type, explain this robustness, notably the presence of too many missing values. As a precautionary principle, we suggest to perform sensitivity analyses where traits are removed one by one or until a certain percentage of removal to assess the robustness of the results (Cooke et al., 2019a; Loiseau et al., 2020; McLean et al., 2018; Mouillot et al., 2014; Pollock et al., 2017). Trait-gap filling through automatic imputation might also be an interesting perspective (Goberna & Verdú, 2016; Johnson et al., 2021; Penone et al., 2014; Schrodt et al., 2015). However, given the way most of these approaches work, this is likely that trait imputations will follow the main trends and the main syndromes and will unlikely generate unique species artificially hidden in the space.

## Species packing in trait space disproportionally increases with species richness

The species packing in trait space, or so-called over-redundancy (Mouillot et al., 2014), provides functional insurance and resilience to ecosystems under disturbances (McLean et al., 2019). This packing can be easily assessed with categorical traits because each unique combination of traits, also called functional entity, is a cluster, so the clusters with a high number of species, or higher than expected under a null model, are considered as over-packed or over-redundant whereas those with few species are vulnerable to biodiversity loss (Mouillot et al., 2014). With continuous traits or a large mix of traits as in our study, the clustering of species remains an arbitrary decision depending on the methods and thresholds used. We chose a clustering method with the lowest number of arbitrary decisions as possible independently of the shape and structure of species distribution in trait space (Rodriguez & Laio, 2014). Surprisingly, this method, despite its attractiveness in other fields (medical and social sciences) and its parsimony (one parameter), has never been applied in ecology and evolution so far.

Using a 'fast search and find of density peaks' algorithm (Rodriguez & Laio, 2014), we show that the number of clusters increases with the number of species when we control for the other factors (Figure 7) but with a strongly saturating relationship (Figure 8a), suggesting that species tend to over-pack into some clusters instead of creating new clusters in species-rich assemblages as shown for reef fishes (Mouillot et al., 2014) or passerine birds (Pigot et al., 2016b). With a slope of 0.41 on the log–log scale, it means that when species richness doubles, the number of clusters only increases by 30%. As a corollary, the richness of the dominant cluster increases with total species richness on a log–log scale with a slope higher than 1 (Figure 8b), suggesting that additional species disproportionally pack into the most speciose cluster. More precisely, two times more species in a given assemblage induces the packing of 2.6 times more species in the dominant cluster. So biodiversity only reinforces the redundancy of the most common traits instead of providing the level of insurance we should expect from species richness only under a random or proportional distribution of species among clusters (Mazel et al., 2014; Mouillot et al., 2014). This remarkable trend is observed for all taxa and ecosystem types.

### The saturating scaling of uniqueness with species richness

The identification of ecological disparity, gaps, distinctiveness or uniqueness in trait spaces is a long-standing issue in ecology and evolution (Bapst et al., 2012; Foote, 1990; Gauzere et al., 2020; Ricklefs, 2005; Violle et al., 2017; Winemiller, 1991). It contributes, for instance, to estimate the level of functional insurance and vulnerability to species extinction (Mouillot et al., 2014) but also to better understand the influence of trait rarity on ecosystem functioning (Maire et al., 2018), to set conservation priorities targeting unique species (Loiseau et al., 2020) and to illuminate the capacity for innovation in clades (Cornwell et al., 2014; Deline et al., 2018; Reeves et al., 2020). Yet there is no consensus on the way to determine which species are isolated enough in trait spaces to be considered as unique species. Among the myriad of clustering algorithms (Xu & Tian, 2015), the method based on fast search and find of density peaks was able to extract unique species in a very intuitive, standard, biodiversity-independent and distribution-free way. We show that the proportion of unique species decreases with species richness (Figure 7) whereas the number of unique species saturates rapidly with species richness (Figure 8c), suggesting that ecological novelty does not scale proportionally with taxonomic diversity but at a much lower rate whatever the kingdom or ecosystem. With a slope of 0.47 on the log–log scale, it means that when species richness doubles, the number of unique species increases by 38%. This result resonates with the saturating link between ecological disparity and species richness across geological periods (Bapst et al., 2012) contrary to predictions from theory on adaptive radiations and ecological speciation (Rundell & Price, 2009). More precisely, some entire lineages remained ecologically conservative throughout the Mesozoic without exploring vacant portions of trait space, and then trait bursts occurred owing to changing abiotic conditions during the Late Jurassic (Reeves et al., 2020). Both adaptive radiations due to species interactions and innovative solutions to face new environments are certainly at play to explain the invariant saturating scaling of ecological uniqueness with species richness.

## CONCLUSIONS

Four take-home messages can be extracted from this analysis. First of all, when no prior selection of traits can be carried out, the minimum dimensionality of trait space is rather large with around three to six dimensions. The success of identifying axes of variation, especially when trait correlations are strong, suggests that the research program of finding major trade-off axes grounded in ecological principles shows more promise than the arbitrary selection and removal of traits. Second, most trait spaces are highly sensitive to trait omission, which thus requires careful thinking about which traits might be overlooked, missed and targeted into the future. Third, there are plenty of unique species, and the success of the clustering approach suggests that we need to pay more attention to how species pack relative to each other in trait space and not only focus on dimensionality reduction of trait spaces. Fourth, the complexity of multicellular organisms from plants to vertebrates or from aquatic to terrestrial species has little influence on the dimensionality, robustness and structure of trait space. Instead, our synthesis suggests that the rate of key functional innovations and the subsequent complexity of trait space are consistent across multicellular clades with multicellularity evolution in plants sharing many features with that leading to animals. Yet these results are based on only 30 datasets and may lack statistical power to detect some effects. Moreover, these results are only valid for the range of dataset characteristics that we used in our analyses so more than 40 species and 10 traits. We obtained different patterns for species-poor assemblages in our simulations, but we are confident that our empirical assessment may embrace most species richness conditions encountered in temperate or tropical assemblages for most taxa when building regional or global species trait space.

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### AUTHOR CONTRIBUTIONS

All authors conceived the ideas and designed the methodology; NL, MG and AA collected the data; DM, NL, MG, NC, MG and AA analysed the data and participated in script development; and DM, NL and AA led the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for submission.

### PEER REVIEW

The peer review history for this article is available at https://publons.com/publon/10.1111/ele.13778.

### OPEN RESEARCH BADGES

This article has earned an Open Materials badge for making publicly available the components of the research methodology needed to reproduce the reported procedure and analysis. All materials are available at: https://github.com/LoiseauN/dimensionality.

## DATA AVAILABILITY STATEMENT

All relevant R codes are available from the GitHub Repository: https://github.com/LoiseauN/dimensionality. Code for the Elbow approach is also available from the GitHub Repository: https://github.com/ahasverus/elbow. All data are available (Table 1).

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## SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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